Prop. Let $V = \{ f : [-1,1] \to \mathbb{R} \mid f \text{ is Continuous} \}$ with equipped an inner product

$$\langle f, g \rangle = \int_{-1}^{1} f(t) \cdot g(t) \, dt$$

Let W be the subspace of odd functions. Find $W^\perp$.

Proof. $f \in W^\perp \iff$ for any odd function g, $\int_{-1}^{1} f(t) g(t) \, dt = 0$

Since for any even function g and any odd f, fg is odd, so $\int_{-1}^{1} fg = 0$.
That is, $W^\perp$ Contains all even function. Moreover, for any $f \in V$,

$$f(x) = \underbrace{\frac{f(x) + f(-x)}{2}}_{\text{even}} + \underbrace{\frac{f(x) - f(-x)}{2}}_{\text{odd}}$$

So, for subset W' of V Consisting by all even functions,

$$V = W + W', \text{ and } W' \subseteq W^\perp.$$

Now, clearly $W \cap W' = \{0\}$, $V = W \oplus W'$.
If $W' \neq W^\perp$, then, for some $h \in W^\perp$, $h \notin W'$. Again, h can be decomposed as

$$h(x) = \underset{\in W'}{\frac{h(x) + h(-x)}{2}} + \underset{\in W}{\frac{h(x) - h(-x)}{2}}$$

Since $W' \subset W^\perp$, $h(x) - \frac{h(x) + h(-x)}{2} \in W^\perp$, and $\in W$.

But, this implies that it is Contained in $W \cap W^\perp$, so it is zero:

If $x \in W \cap W^\perp$, then $\langle x, x \rangle = 0$, or $x = 0$.

Thus, $h(x) - \frac{h(x) + h(-x)}{2} = \frac{h(x) - h(-x)}{2} = 0$, So $h(x) = h(-x)$, even. Contradicts to $h \notin W'$.

Now, we can Construct a Projection of V onto W as:

$$P : V \to V : f(x) \mapsto \frac{f(x) - f(-x)}{2}$$